

MADHAVA PRASAD K G

Thiruvavarur, India | +91 6384360663 | prasadmadhav44@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Software engineer and full-stack developer specializing in multi-agent LLM applications, REST API design, and modern web development. Built and deployed production-grade systems using Python, Flask, Google ADK, React.js, Node.js, and MongoDB — including an AI-powered multi-agent financial analyst and a full-stack e-commerce platform. Certified in MERN stack development, frontend technologies (React, TypeScript, Next.js), backend engineering (Node.js, Express.js, MongoDB, AWS), AI agents, and data analytics. Seeking roles in full-stack software engineering.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Bootstrap 5, Vite, React Router, React Hook Form

Backend: Flask, Node.js, Express.js, REST APIs

AI / GenAI: Google ADK, Prompt Engineering, Multi-agent Orchestration, LangChain

Databases: MongoDB, Pandas

DevOps & Tools: Git, GitHub Actions, AWS, Docker, Vercel, Firebase, Render.com, Zod, yfinance

PROJECTS

Ecommerce Store — Full-Stack React Shopping Platform | 2026

React 19 · Vite 7 · Tailwind CSS · React Router v7 · React Hook Form · Zod · Context API · JavaScript ES6+

- Built a production-quality full-stack e-commerce platform featuring product catalog with search, category filtering, and multi-criteria sorting; implemented global cart and wishlist management via React Context API with localStorage persistence across sessions.
- Engineered form validation pipeline using React Hook Form and Zod schema validation; implemented lazy-loaded routes with React Router v7 Suspense boundaries and custom React hooks for REST API data fetching, reducing initial bundle size for deferred components.
- Architected reusable component library (ProductCard, Cart, Checkout, PageHeader, EmptyState, Button, Toast) across 10+ pages with shimmer skeleton loading states and dismissible toast notifications; styled fully responsive layout using Tailwind CSS utility classes.
- Structured scalable codebase with Context API global state, utility helpers, and custom hook abstractions — demonstrating production-grade code organisation, clean separation of concerns, and maintainable software architecture.

AI-Powered Multi-Agent Financial Analyst | 2025

Python · Google ADK · Gemini 1.5 Pro · Flask · yfinance · Bootstrap 5 · Render.com · REST API

- Architected a 3-agent LLM orchestration pipeline (Research Agent → Planner Agent → Risk Evaluator Agent) using Google ADK and Gemini 1.5 Pro, enabling autonomous equity analysis across 10+ stock symbols per session with sub-5s end-to-end latency [ESTIMATED].
- Engineered structured prompt chains with Gemini 1.5 Pro for risk-tiered, explainable investment recommendations; implemented JSON schema output validation against live yfinance data to reduce hallucination on price-sensitive financial outputs.
- Deployed Flask REST API + Bootstrap 5 frontend to Render.com with CORS, rate limiting, session-based state management, and security headers; designed modular agent JSON handoff schemas enabling independent agent swapping without pipeline disruption.

Wireless Patient Alert System — Embedded Real-Time Communication | 2024

Arduino Uno · C++ · 433MHz RF Modules · Interrupt-driven event logic · LCD · Arduino IDE

- Designed and prototyped a real-time wireless alert system across 3 hospital ward nodes using Arduino Uno and 433MHz RF modules; implemented interrupt-driven C++ event logic with LCD status display and buzzer escalation alerts.
- Validated 71% reduction in alert response time (7s → <2s baseline) across 50 controlled test cycles — demonstrating measurable performance optimisation and systems engineering under real hardware constraints.

LEADERSHIP

- TinyML Workshop Organiser:** Led a university-level workshop on TinyML for edge devices with 150+ student participants; managed speaker coordination, hardware lab setup, and live Q&A sessions throughout the event.
- Cybersecurity Workshop Participant:** Completed hands-on penetration testing fundamentals using Kali Linux in a 200+ student workshop environment, gaining foundational knowledge in network security and vulnerability analysis.

CERTIFICATIONS

5-Day AI Agents Intensive Course with Google - Google × Kaggle (Dec 2025) | **MERN Full Stack Development** - Code Galatta

Frontend Dev Training - Code Galatta · React, TypeScript, Next.js | **Backend Dev Training** - Code Galatta · Node.js, Express.js, MongoDB, AWS

Deloitte Data Analytics Simulation - Forage (June 2026) | **Prompt Engineering for ChatGPT** - Great Learning (Nov 2024)

Introduction to Artificial Intelligence - Great Learning (Nov 2024)

EDUCATION

B.Tech, Electronics and Communication Engineering - SASTRA Deemed to Be University | 2022 - 2026 | CGPA: 7.1351 / 10

HSC (Class 12): 88.67% | **SSLC (Class 10):** 79.6% - Sri Shanmuka Matric Hr. Sec. School